

Equipe UFRJ

Minerva Baja's Project Report

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1st Student Challenge
SAE Brasil & BALLARD Student H2 Challenge 2020

1. INTRODUCTION

The environmental impacts caused by vehicles using fossil and non-renewable fuels have been a topic of much debate nowadays. This concern is mainly justified by the fact that such models are the majority in the total number of vehicles in the world. In addition, conventional vehicles that use petroleum-based fuels has, as product, greenhouse gases that contribute to the rise in the average temperature of the planet. Besides that, these vehicles use, as their main form of energy converter, internal combustion engines that have a low energy efficiency rate, floating around 25%. Thus, it is necessary to develop new technologies to counter such problems.

From this perspective, the fuel cell technology was developed. This device is responsible to convert the chemical energy in the fuel molecules into electrical energy, and this process has a high-efficiency rate. In this challenge promoted by SAE in association with Ballard, we'll have as focus a hydrogen-based fuel cell, in which the energy is obtained through the oxidation reaction of that substance, which will have water as a product.

2. DEVELOPMENT

This document describes all the stages of the Minerva Baja Team's project about the development of a fuel cell hybrid vehicle, utilizing a Baja SAE as platform. SAE Brazil idealized this project in partnership with Ballard, which will be donating the Fuel Cell Stacks, in the form of a Hydrogen Challenge that will be a competition between Formula and Baja SAE Teams.

In order to develop a fuel cell hybrid powertrain into the Baja vehicle platform, some adjustments were necessary in the existing project. Besides that, new components were included in the vehicle and the way of integrating them into the system was the main challenge of this project.

According to [24] the use of a fuel cell alongside a battery pack allows the surpassing of some FC inherent limitations when using it to power a vehicle. As mentioned by Wang and Nehir [25], a FC is not capable to respond quickly to accelerations. Furthermore, its efficiency is poor when working in lower power ranges [26]. Thus, the use of a battery pack alongside the Ballard FC1020 ACS allows a better performance in accelerations and in low FC efficiency ranges.

2.1 SYSTEM MANAGEMENT –

In order to keep all the electric components working in a safe and optimized way, it is mandatory the use of a system management unit which will control the components to achieve a desired result.

With that in mind, the energy flow of the battery bank, fuel cell, and electric motor can be determined by the control strategy configured in the vehicle controller. The vehicle's power system management strategy must ensure three conditions [26]:

• The power of the electric motor must always follow the vehicle's power demand;

• The fuel cell operates only in its high-efficiency range;

• The charge status of the battery pack is maintained between its minimum and maximum state of charge (SOC), those being defined later on.

The basic structure of the system management is composed by two different systems. The first one is the monitoring system, which will gather and analyze all the data provided by the sensors in the system. This data will be the input of the control system that will actuate through commands in a programed controller to regulate the behavior of the system devices.

One of the main components used for this purpose is the motor inverter, which has an integrated PID (Proportional-Integral-Derivative) controller with programmable parameters to control the induction motor. For the purpose of the project, we chose the WEG CVW 300 illustrated in the appendix **A11**

that has the proper specification to power and control the chosen motor with a 9.2kW power.

For the purpose of the Hydrogen Challenge, the team had to limit the nominal input power in the motor to 4.2 kW. To achieve that goal, it is necessary first to understand the basics of the inverter. The input power in the inverter will be equal to the output, despite some losses in the conversion of the DC to AC [8]. From there, we have to equalize the input power to the active power equation (ACE) of the output.

$$ACE = \sqrt{3} * V_{Motor\ nominal} * I_{Motor\ current} * \cos\theta$$

In which $\cos\theta$ is the motor's power factor in 100% of the charge (0.86 in the chosen motor).

$$4200W = \sqrt{3} * 36V * I * 0.86 \qquad I = 78.3A$$

From this value we can program the inverter parameter to limit the motor current from its nominal value.

Also, the nominal battery voltage is around 51.1V, with that parameter the current drawn from the system can be calculated, and it is around 82.3A, in the nominal state. That current will be drawn from both battery and fuel cell, once they are connected in parallel.

$$P = V * I \qquad 4200W = 51V * I \qquad I = 82.3\text{A}$$

On this perspective, once the voltage level of the fuel cell is different from the battery pack, urges the necessity of a DC/DC converter, which in this case will boost the stack voltage, which is lower than the battery pack nominal voltage. With this in mind, it is necessary to control the converter in order to limit the stack voltage and current.

In other hand, there is a monitoring system, which will collect the data from the devices in order to keep them inside their specific work range. In Table 1 it is shown all the sensors present in the system and the specified range of the device they are measuring.

Table 1: Sensors, its functions and control parameters

Sensor	Function	Control Parameter
Pressure Sensor 1	Measure the pressure in the cylinder to estimate autonomy	0.36 barg – 153 Bar
Pressure sensor 2	Measure the pressure in the anode of the stack	0.16 barg –0.56 barg 0.36 barg Optimal
Thermistor	Measure the stack temperature	6 ° - 75 °
Voltage Sensor 1	Measure the voltage in the stack terminals	0V – 48.3V
Current Sensor 1	Measure the current drawn by the stack	0A – 78A
Voltage Sensor 2	Measure the voltage in the battery pack terminals	35V – 58,8V
Current Sensor 2	Measure the current drawn by the battery pack	0A – 175A

The data obtained from these sensors will serve as input for our control system, which will read and actuate based on that data. On the controller programming, if one of those device's parameters stay out of their respective work ranges, the control system shall act in order to avoid damage to the powertrain devices.

The vehicle may work in three different modes, each with a distinct control strategy involving the Fuel Cell (FC), the Battery Pack (BP) and the Induction Motor (IM):

1. Static mode: None of the two energy sources passes energy through to the IM;

2. Braking mode: FC is deactivated and BP is charged by the energy coming from the regenerative braking through the IM;

3. Traction mode with the BP in mode Charge Sustain (as seen on ?):

If the power required by the IM is between the FC minimum activation power and its maximum power and the BP doesn't need to be recharged, the FC acts alone to provide the required power by the IM. In case the BP needs to be recharged, the FC works on its full power and the excess energy not used by the IM charges the BP

If the power required by the IM is higher than the FC's maximum power, the FC and the BP are both triggered. The FC provides its maximum power and the BP provides the extra amount necessary to meet the IM requirement.

In case the power asked by the IM is lower than the minimum FC activation power and the BP doesn't need to be charged, the BP works alone to provide power to the IM. In the scenario that the BP actually needs to be recharged, the FC works on its maximum power and the amount not used by the IM is used to charge the BP.

One of the main components we need to monitor is the fuel cell, having in mind that if one of its parameters is working out of its range there may be permanent damage to the stack. For that reason, there are some procedures to follow both during the start-up and shutdown that will make sure that the correct use of the stack is made. Such procedures must be

programmed with the inputs of the monitoring system to perform correctly and perform the steps.

Start-up procedure logic – These steps must be followed in the program of the controller and in this order to ensure a proper startup of the fuel cell:

1. Turn on fan – which will both provide the oxygen to the stack internal reaction and control its temperature. It will also make sure to dilute any residual hydrogen in the air flow.

2. Open the supply valve and purge valve – Which will provide fuel to the stack and push air out.

3. Close the load contactor – With this action, current can be safely required from the stack

Shutdown Procedure logic – In order to safely manage the fuel cell when current will no longer be required from the stack, those steps shall be followed in controller program:

1. The first step is removing load from the stack and close the load contactor. This is very important because if fuel flow is closed before the load, the stack can starve, causing permanent damage.

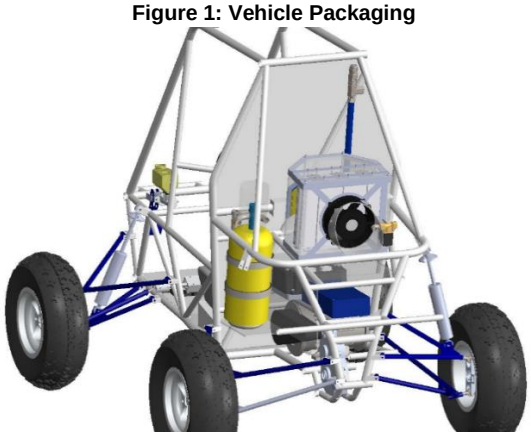
2. The supply valve can now be closed. Basically removing the voltage from it, given that the used valve is normally closed.

3. At last, oxygen supply can now be ceased, and for it, the fan is shut off.

2.2 FUEL CELL HYBRID POWERTRAIN –

Our main objectives in this section was to develop the packaging of all the components and make sure that they were following all of its safety rules. Besides that, each component was chosen by its own specifications in order to work properly in the whole system. Following the block diagram shown in appendix **A12**, we can notice the main components of the powertrain system.

Figure 1: Vehicle Packaging



Hydrogen cylinder – Once working with a gas-powered device like the fuel cell, there is a necessity of some storage for this gas to keep the constant feeding of the fuel cell. With this in mind, a tank was to fulfill that necessity, in which the hydrogen will be stored at high pressure (153 Bar of working pressure) due to the autonomy of the vehicle. The partner company Air Products will provide that cylinder to the teams of the challenge; thus, this project was based on that model.

In order to keep the vehicle safe for performance, we followed the rules set by the challenge organization [21]. Among the 3 options for the hydrogen tank placement, we chose the "Vertical mounting behind the firewall", slightly leaned backwards, as it is the one that best adapts to the layout of the powertrain system designed for the car. We have positioned the cradle (seen on Appendix **A17** and in Figure 1) so that there is enough space for the installation and removal of the tank, as well as ensuring a distance of 80mm from the filling plug to the car frame, as defined by the rule. We used 1020 steel (1-inch diameter) for the cradle piping for easy manufacturing, since the team already has this material. The cradle attachment to the vehicle structure will be done through MIG welding, to attach the belts to the cradle, a lap will be made on the tube and spot welded with at least 100 mm of overlap will be used. In order to restrict any possible displacement of the belts, considering that the cradle is positioned upwards, steel tabs have been welded to the cradle just below the belts. For safety reasons, a shield composed of polycarbonate and rubber layers was placed between the cylinder and the main sources of heat and possible electric sparks (electrical motor, battery, inverter) in order to thermally and electrically isolate the cylinder.

Main Cutoff valve – Due to the need for a manually operated valve in order to separate and cut the hydrogen flow, we chose a model that was easy to find, financially viable, and mainly that met all the specifications of the rule since this is an essential item for the integrity of the fuel cell stack. Facing several options, we opted for the White Martins valve 40000606. In addition, we carefully evaluated the placement of this valve on the vehicle so that it is easy to use and accident-proof.

Pressure regulation valve – Since the hydrogen is in high pressure inside the tank, we must have some component to adapt the high working pressure of the cylinder to the low working pressure of the fuel cell. Therefore, we use a regulating valve for this pressure control. For our application, we must take into account the degree of precision in adjusting this control, as well as a double stage method for adjusting the pressure. The cylinder working pressure that we will use is about 153 bar, and from there, it is necessary to decrease this value to about 0.36 Barg, which is the fuel cell optimal pressure to work. In this perspective, White Martins' model LFD-15 **A7** was chosen, as it provides the characteristics required.

Control valve – In order to electronically control the hydrogen flow towards the fuel cell, we decided to use a solenoid valve applied to the low-pressure system, controlled by both the control system and the vehicle panic buttons. The choice of a solenoid valve is due to its fast and reliable acting, which works through electric current, ensuring a quick interruption of the hydrogen flow in cases of emergency and thus ensuring driver safety. Furthermore, it protects the stack against starvation, avoiding severe damage to the fuel cells.

The chosen valve operates in a normally closed binary on-off state — in other words, it remains closed in the absence of current — and its state is only changed during the startup and shutdown procedures. In addition, it does meet the draft rules and specifications in [21]: suitable for use with 99.95% pure hydrogen, rated for a minimum pressure of 200 psig (13,8 barg). It also meets IP65 protection degree (which indicates resistance against dust and water jets) and has an EPDM seal.

As determined, we intend to fasten the valve to the vehicle's fairing, using grade 12 bolts and self-locking nuts, with enough clearance from surrounding parts.

Pressure relief valve – For safety reasons, the use of a valve to avoid any overpressure in the low-pressure system is recommendable. This is crucial to maintain the fuel cell anode in a safe range of pressure in case of any other safety device failure.

Once hydrogen may flow out from the valve, as a further safety measure, we positioned it at the upper left side of the Rear Roll Hoop in a way that the valve points up and away from the cockpit and critical components of the vehicle.

Fuel Cell – It is the main component of the project and it is from there that all systems, adaptations, and innovations are based. The Fuel Cell used is FCgen®-1020ACS, part Number 5127057, containing 46 cells, manufactured and supplied by Ballard Power Systems. The team used the Fuel Cell Integration Manual [22] provided by Ballard as a reference for cable installation, handling, and safety procedures, in addition to the competition rules and regulations [21].

When it comes to the Fuel Cell Stack positioning and orientation, it was assembled as a floating electrical equipment, with both anode parts facing downwards and the outlet lower than the inlet, one of the arrangements recommended by Ballard. Around the equipment a custom-made enclosure was designed with the following objectives in mind:

1. Keep the stack in the planned orientation, ensuring proper operation;

2. Provide ducting to allow the fan to ensure effectively air flow through the stack, so that the temperature stays inside the optimum range, as shown in [22], and the cell does not starve;

3. Prevent unintended or unauthorized contact with the stack by the end-user or others;

Finally, polycarbonate sheets are used to ensure that conductive materials never come in contact with the cells or the bus plates. However, for this application, aiming higher structural rigidity to the enclosure (as seen on **A21**), it became necessary to implement a frame made out of aluminum. It is mounted on the external part, so the polycarbonate sheets are isolating any conductive materials from the Stack, preventing short circuits, as recommended by [22]. Furthermore, rubber shock mounts shall reduce vibrations in the stack. As for the placement of the stack, it is located behind the fire-wall, completely contained inside the vehicle's fairing and above 50mm of battery, as established by the rule [21]. With that in mind, the tank shield was extended so that the whole hydrogen system is completely isolated from the electrical system, including the Stack, as shown in Figure 1

Purge Valve – This valve is electronically controlled and it works on pulses where the purge intervals are determined by a function that depends on the current required by the fuel cell, since with a higher current the hydrogen consumption will also increase, and therefore there will be an increase in the purge frequency.

DC/DC Converter – The DC/DC converter will be used to increase the voltage that comes from the Fuel Cell to supply the battery dedicated to the traction motor. Basically, a current generated by a DC source arrives in an inductor that generates a magnetic field that replicates this current whenever when passes through it. The cyclic characteristic of the circuit is assured by a switch. Finally, a diode coupled to a capacitor concentrates the final voltage in this capacitor, releasing the current flow.

Due to commercial limitations, an association of three equal converters in parallel is used to supply this function. The converters utilized withstand 10V to 60V of input voltage and convert into a range of 12V to 90V at the output. They operate on a current up to 30A, resulting in output power of 1500W.

This way, it is possible to work with the 75A current coming from the fuel cell, providing the vehicle's specifications.

BMS – With the connections in series made in the lithium-ion battery pack, the batteries become unbalanced. When a charge is applied to several cells connected in series, despite the charge current being the same in all cells, the voltages will be different in each cell causing its imbalance. The greater the imbalance of the cells in a pack, the higher the loss of capacity after a few cycles of loading and unloading. To increase performance, the Battery management system (BMS) is used to balance all lithium-ion batteries in the pack. Whenever the first cell reaches the minimum cut-off voltage, the BMS will stop the battery discharge. It protects the battery cells from being discharged beyond the minimum which would cause irreparable damage. It also cuts off battery power if the battery charge exceeds a certain threshold, in addition to protecting it from possible short circuits. With the high charge and discharge rate, lithium-ion batteries overheat, especially during recharging while braking (due to its regenerative nature), the frequency of the rotor is greater than the frequency of the stator, causing a reverse flow of energy from the rotor to the stator. The motor then works as a generator, injecting this energy into the DC bus of the inverter and directed to the batteries.

In this situation, the BMS has a fundamental role in continuously monitoring the temperature of the batteries and comes into action when the temperature is outside the desired range. The life of a lithium-ion battery depends largely on its operating conditions, too high or low temperatures shorten the life of the battery. Therefore, the BMS prevents it from overheating during its use, being a fundamental component for the performance of the hybrid powertrain.

As the objective is to supply a battery of 51.1V and capacity of 14Ah, a BMS for Lithium-Ion battery that works in 100A for 7 to 16 cells. This way, the device will be sufficient to manage what is required by the vehicle.

Battery – In the Baja hybrid vehicle 40% of the traction power comes from the lithium-ion battery pack. This package consists of 98 3.65V, 2000 mAh cells ordered in a series of 14 groups of 7 cells in parallel, guaranteeing a 51.1V nominal voltage, which is required by the inverter for powering the traction motor. The choice for this battery type is motivated by the need for high energy density and light weight, in addition to its advantages compared to batteries with traditional technology, lithium ion batteries charge faster, their charge lasts longer due to their relatively low self-discharge, it is less than half the self-discharge of NiCd and NiMh batteries. Low Maintenance, no periodic flushing is required, in addition to having no memory. This indicates that, even if the cells are not loaded to 100%, they will not "addict" increasing the vehicle's durability and life. The high energy density allows the potential for even greater capacities, increasing battery life even in a smaller structure.

Inverter – Since we use an AC motor in the project, we need an inverter to convert direct current from the batteries and the fuel cell to Alternating current. The inverter is essential to the powertrain system, mainly for its ability to control engine behavior, for its ability to offer precise variable control of the speed of a motor by providing variable voltage, current and frequency, with those parameters being defined via a feedback loop where the driver will require different response from the car. Thus, the driver feeds the system his wishes by interacting with brake and throttle pedals, coupled with the DS1901 TPS (Throttle Position Sensor), widely available on the market, normally used on popular cars, with a cost of around R$70. This inverter is designed to work with batteries from 24 up to 72V and currents of 100, 200 and 400A for up to 2 min, having a nominal current of 200A, torque and speed control as far as 0 rpm (vector control with encoder) and static accuracy of up to 0.01 % of nominal rotation. We positioned the inverter in a spot with a considerable air flow in order to provide better cooling, so that we keep the surrounding temperature within the recommended limits, according to its manual [20], and keep all connections as short as possible to minimize noise and improve electromagnetic compatibility.

Motor – The motor is a crucial component in the car, it converts electrical energy provided by the PEM fuel cell and the battery, transforming it into mechanical energy that is used to power the vehicle. For budget reasons we are opting to use a three-phase AC induction motor the team already owns with the following specifications: 9,2 kW power, 198 A nominal current and 36 V nominal tension. Its power though, is being limited to a nominal of 4,2 kW to match the challenge requirements [21]. The cradle used in our previous BAJA model is being adapted to fit the electric motor. One of the adaptations is the positioning of the cradle parallel to the ground instead of a 15% angle from the former combustion engine, with that adaptation we aim to lower the center of gravity of the car, making it more stable.

2.3 TRANSMISSION

Abstract – The job of the mechanical transmission is, in sum, in all applications, to pass on torque from the engine to the wheels and also transform this power to a configuration that is usable, preferably optimized to each circumstance, maximizing efficiency and longitudinal performance. In this challenge we will be using and adapting a Baja SAE platform, with a transmission that used to consist of a Gaged GX9 Continuously Variable Transmission (CVT), with a reduction that varies from 0.9:1 up to 3.85:1, a fixed reduction gearbox, with two straight teeth gear reductions in series, adding up to a 6,85:1 relation, calculated through a resistive forces model based on [3]. All gearbox internal components were sized through the GearPro software (based on [4]), following the AGMA Standards. Such program was developed by a fellow UFRJ student with the aim of minimizing the time spent during sizing and modeling phases. After some thought we had to decide the gear material, and between SAE 8620 and SAE 4320 we chose the latter. Even though its cost is 75% higher, we picked him for its higher yield stress, which allowed us to reduce mass and, consequently, less inertia for the same safety coefficient, resulting in a slight improvement in the vehicle acceleration. We also quenched, tempered and went through a liquid carburizing process on the teeth with a depth between 0,6mm e 0,9mm.

To verify the heat treatments done, which by [7] would result in superficial hardness in the range of 58 HRC and 60 HRC, we done hardness tests on the gears and obtained a result of 60HCR, as expected. In sum, we got a fixed reduction gear box with a safety coefficient of 1.24, respecting the 1.2 value stipulated for this project. In comparison to the previous gear box we ended up with a 40% (3.6kg) and 35% (3000kg*mm²) reduction in mass and moment of inertia, respectively. Furthermore, we had a 14% reduction in the safety coefficient, justified by the knowledge improvement by the team on the workloads.

After the reductions, we use a constant-velocity joint originally from the Renault Kangoo (widely available in the market) to finally transmit torque to the wheels, which, via the tires and its friction with the terrain, will provide kinetic energy to the whole vehicle. Given that, we had to come up with a solution that best suited our needs for the adaptation of the hybrid electric powertrain into our platform, heavily considering practicality and cost efficiency without a heavy loss in performance.

Adaptation – We beforehand removed the components of the old transmission that are not needed in this new configuration, those being: the Internal Combustion Engine (ICE), Gasoline Tank, and CVT, which will not be necessary given the linear characteristic of the torque curve of the Electric Induction Motor (IM) inside its desired workable range.

Since we strongly value factors such as cost cutting and feasibility, it was decided that we would keep the same fixed reduction gearbox in the package, allowing the CV Joints to be held in place without the need of other components, and furthermore taking away the need of dimensioning and machining new components to replace our current gearbox. Then, after remodeling the motor cradle to fit the IM, we finally had to dimension and model the reduction between the motor itself and the gearbox input shaft. In that sense, we had to

initially decide between a chain and a gear reduction. And we did that with the help of a decision matrix, where we defined factors and their respective weights, as seen on Table 2.

Table 2: Primary reduction's Decision Matrix

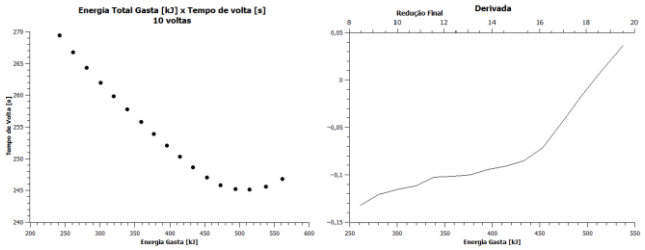
Factors	Weight	Chain	Gear
Cost	3	x	
Ease to obtain	3	x	
Simplicity	2	x	
Efficiency	3		x
Occupied Space	1		x
Total		8	4

Now it became essential to size the sprockets and the chain, but before it is essential to define our objectives and parameters. Initially we ran a preliminary simulation using the software OptimumLap, having in mind Lap Time and Energy Spent as objectives to be optimized. In the software we designed the reference racetrack for virtual performance evaluation and fed as input data parameters such as:

- Car Weight: 3150N
- Rolling Resistance: 0,03 (Obtained through an experiment, as shown in appendix **A1**, which consisted of half-vehicle structure attached to a car running in a constant speed through a rope with a Load Cell in between them. In a manner that the load cell would register the resistance forces acting on the structure. Furthermore, the rolling resistance was also validated by the same coast down test)
- Engine Torque Curve (adapted to fulfill the power limitation requirement). Appendix **A2**
- Drag Coefficient and vehicle frontal area: 0,85; 0,88m². Those values were obtained through a Solidworks CFD analysis, as can be seen in appendix **A3** and validated through a coast down test, based on [5] and [6], the latter following the SAE J1263 standard. The CVT belt and the intermediate gearbox shaft were removed in order to avoid losses by mechanical resistance and viscous friction, respectively, once the theoretical resistive forces model considers only rolling resistance and aerodynamic drag. The prototype was accelerated up to 70km/h and then uncoupled. For data acquisition we used a Hall effect sensor, with an uncertainty of 1km/h and average deacceleration measured in intervals up to 10km/h. The test was made in both directions with a total of 10 runs, to minimize variations caused by inclination. The image shown in **A4** shows the experimental points plotted and adjusted to a quadratic function with a null linear coefficient along the theoretical curve. The discrepancy is due to imprecisions in regards to ambient conditions as demanded by the standard.

Using that data, we iterated a 10-lap run (to minimize the influence of the acceleration lap in the results) varying the final gear ratio each time. As a result, we got a Final Ratio x Energy Spent [kJ] curve (**A5**) and a Final Ratio x Lap time [s] curve (**A6**), from which of those we extracted data with the goal of optimizing our final reduction choice also having in mind fuel saving and energy efficiency. Our line of thought was: We need to reduce our total energy consumption without compromising our longitudinal performance, translated in this analysis as average Lap Time. Having that data output from OptimumLap we plotted an Energy Spent [kJ] x Lap Time [s] graph and then it's differentiated version, as seen on Figure 2 with the help of QtiPlot, a plotting software commonly used in our academic environment.

Figure 2: Energy Spent [kJ] x Lap Time [s] graph and its differentiated counterpart



On that final graph, we can clearly read a change in behavior on regards to the tradeoff between a lap time gain and a lightening of the total energy spent. In other words, for final gear ratios bigger than that equivalent to that change in behavior we gain more lap time than we are able to accept in order to increase our fuel efficiency.

Having that under account we chose a final gear ratio of 13.7:1, which translated into a relation of 2:1 on the primary reduction (chain and sprocket). Ultimately it became essential to size the sprockets and the chain, with the aid of [1] , and having as input data the following:

- Torque provided by the engine: 14.8 Nm
- Nominal Power: 5.5 hp (4,2kW cap)
- Nominal rotation: 2710rpm (considering the power cap)

After some calculation iterations, the distance between shafts was defined as 285.6mm and we ended up with the optimized choice of using (following the catalog [2]):

Table 3: Chain and Sprocket Data and Cost

1st Sprocket	23 teeth	Din 06C-01 (ASA35)	R$24,32
2nd Sprocket	46 teeth		R$72,45
Chain	92 links	ASA A35	R$40,85

It's crucial to add that those Sprockets and Chains are non-expensive and widely available in the market, with an estimated total cost of R$300,00 including chain, sprockets, shipping costs and machining (keyways and gear hubs) with one of our partner firms.

Finally, we did an estimate of the final weight of the vehicle, arround 3150N, and with all the changes in the powertrain done, considering the reference track for virtual evaluation provided by SAE, and only the 4.2kW power (not the possible “boost” up to 7.5kW) the vehicle hits a theoretical maximum speed of 34,97km/h, and a maximum longitudinal acceleration on 0,92m/s².

2.4 BRAKES / COCKPIT

Brakes - In order to achieve comfort and safety objectives, the following goals are defined based on the values recommended by [9] and [10]:

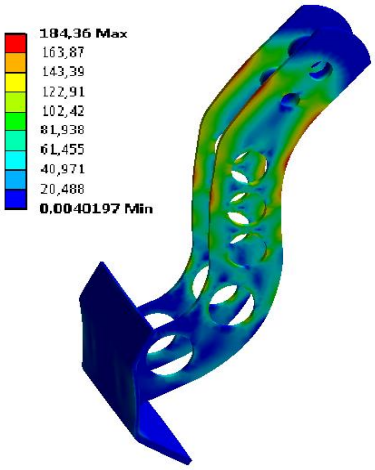
- Minimum average deceleration of 5,8m/s2;
- Maximum braking distance of 27m;
- Pedal distance travel between 100mm and 150mm;
- Maximum work of 500N on pedal.

That done, the distribution of the braking forces is chosen according to the dynamic model presented on [10]. The value is defined to lead to simultaneous axles braking with a deceleration of 0.7g, obtaining a stable breaking in all dirt and mud strip. From this, the friction coefficient must be higher than the one found experimentally (maximum of 0.66) due to dirt variability, as higher values can be found in other regions. As a result, theoretical minimum average deceleration of 7,35 m/s² and braking distance of 20.25m are obtained, both values for asphalt as required in [9].

Aiming for a minimum safety coefficient of 1,2, a structural analysis is made on ANSYS considering abrupt braking on asphalt (largest friction coefficient obtained experimentally), and based on [10]. In this analysis, a normal force of 3,6x106 N and momentum of 312 Nm are applied at the front disk on the contact area with the brake pad, with fixed supports on the disk mountings. From the stress distributions, different materials for the disk are compared, and according to appendix **A10**, SAE 1045 steel is chosen mainly due to the higher flow limit, providing a security factor of 1,25 on the front disk and mass of 0.296 kg.

For the pedal design, a ratio of 8,31 is chosen due to the 273 N/g force gain, as this value is inside the “very good” range. In addition to a maximum travel distance course of 135 mm, both results are according to [10]and meeting the system goals. Moving to the structural analysis, a minimum safety coefficient of 2,5 is the goal — a higher value compared to the other components as it’s a main safety component of the system — the maximum force applied is 823N as quoted on [10]. For standard conditions of structural analysis, cylindrical joints are used on the pivot point and push rod fixation, with the force applied evenly distributed on the pedal. For the material, SAE 1045 Steel is chosen due to the low cost and less susceptibility to the effects of thermally affected zones. In conclusion, a safety coefficient of 2,88 and mass of 250g (a reduction of 33%) is achieved.

Figure 3: Stress distribution on the brake pedal



With the objective of validating the utilized models, tests of the braking distance are made, in which velocity is measured through the system presented on the section relative to electronics. Results are exposed on table 4. An error of about 13% is noticeable, which justifies itself mainly by the disregard of system pressure loss and response time, both from driver and system. Furthermore, a temperature test is made with an infrared sensor to validate the thermal analysis model utilized. After, pressure and velocity values are used in an ANSYS analysis to calculate the power dissipated. The objective is to compare the theoretical and experimental results, leading to the results present in figure 4. Acceptable error is shown, considering that the stabilization system temperature is far from the fading according to [10].

Figure 4: Theoretical and experimental results for the front brake disk temperature x time

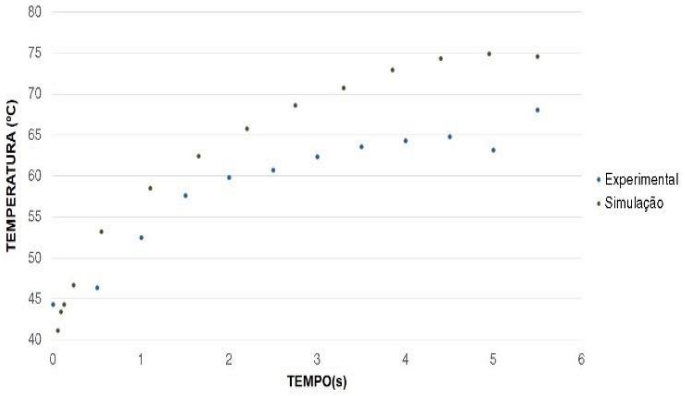


Table 4: Theoretical and experimental braking distance

Test number	Distance(m)	Test Number	Distance(m)
1	23,6	5	23,4
2	22,7	6	22,4
3	23,5	7	22,8
4	22,8	8	23,4

Tires - Considering the project’s dependency of tire behavior, several experimental analyses are made on the available

tires in order to dimension the components of the prototype. The vertical stiffness from the tires is obtained in a universal testing machine, ranging from different tire pressures, while the coefficient of friction is obtained placing the previous prototype parked with the brake system triggered. From that, the vehicle is pushed to imminent movement, with the process being repeated for different soils, different filling pressures, and vertical loads, as the maximum force is measured through a load cell. Part of the results are summarized on Table 5.

Table 5: Experimental coefficients of tire-soil friction

Terrain	Coefficient of friction
Asphalt	0,86
Wet asphalt	0,76
Cobblestone road	0,75
Dirt	0,43-0,66
Mud	0,31-0,39

Furthermore, it's necessary to evaluate the rear tires' tractive capacity. This procedure is done by attaching one side of the load cell as the prototype pulls the other side until the tires slip. The variables are the same as the ones from the test previously mentioned. On the other hand, it's necessary to develop a structure to obtain the lateral forces curves and rolling resistance coefficient, as shown on [11], in which the vertical load and the slip angle of the tires vary. The experiment is shown in Appendix **A1** and **A8**.

A greater range of data is obtained, and from these data, adjustments are made by the Pacejka magic formula from 97, and through the similarity method presented in [3], tire modeling for the pure lateral force situation is achieved (figure 5). Some results are summarized in Table 6. From the obtained results, Sun-F was chosen for the front axle, mainly due to his lateral stiffness and its directional carnations, while for the rear axle Dunlop KT401 was selected, aiming for a greater tractive capacity.

Figure 5: Lateral force x Slip angle

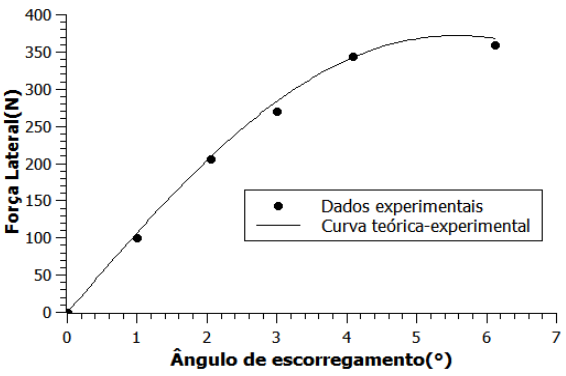


Table 6: Experimental data from different tires

	Sun-F	ITP Mud Lite AT	KT401
Dimensions	22x7-10	22x8-10	23x8-10
Pressure(psi)	9	9	9
Vertical stiffness(KN/m)	70	73,2	64,9
Tractive capability(N)	1310	1458	1532
Lateral stiffness(N/°)	107	131	98

Cockpit – In decoherence of the new SIM shape, the internal area of the chassis increases 4,25%, and hence the distance between the points referenced as P in the RATBSB [23] increase 14mm. This increase lead to the solution of recurrent problems for taller drivers as difficulty to enter the car, and knees hitting the SIM as the car trespasses a rough terrain.

In order to have the maximal force application at the brake pedal, as referenced in [17], the angles of seat and backrest are defined as 15° and 100° respectively. From the backrest inclination, the length of the lumbar support is defined as 5cm, based on [17]. The backrest and seat also are designed with the goal of offering more lateral support for the driver.

In addition, the position of the steering wheel is changed. This alteration positions the steering wheel closer to the pilot and decreases radial and ulnar deviation. The deviation angles reduce from -30° to -9°(**A9**).

2.5 SUSPENSION AND STEERING

The suspension and steering systems were designed to meet the following requirements:

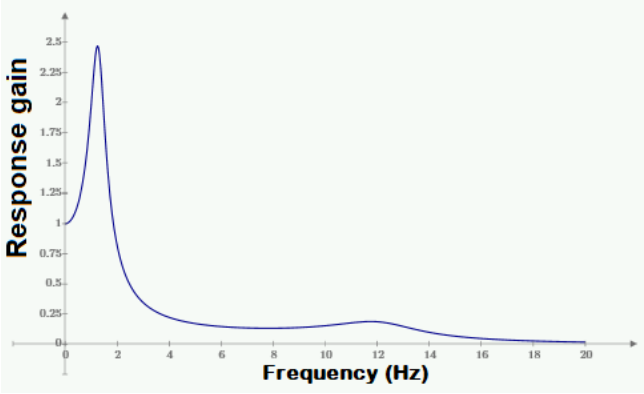
- Low sprung mass response gain to road displacement inputs in high frequencies
- Understeer gradient evaluated at 0.6g higher than -3.0 º/g
- Structural reliability

The suspension design began studying the vertical dynamics through the one-quarter car model with two degrees of freedom. First the values of the natural frequencies, 1.35 Hz in the front suspension and 1.45 Hz in the rear, were set based on past experiences [11], the rear value being slightly higher in compliance with Olley's criteria[18].

The damping ratios were then chosen so as to keep the sprung mass response gain low in high frequencies. The criteria used to determine the values assesses ride quality and road holding performance, as introduced in [12].

Compression/extension values of 0.20/0.30 and 0.25/0.35 were chosen for the front and rear suspensions, respectively, the ratio being lower in compression due to the higher speed in bump, as recommended in [12]. The front suspension sprung mass response to road displacement inputs, for instance, is shown in Figure 6. Despite the peak in amplitude for values around the natural frequency, the response is halved at 2.5 Hz and beyond 3.5 Hz it stays below a quarter of the input. The result is a smooth ride and thus increased driver comfort.

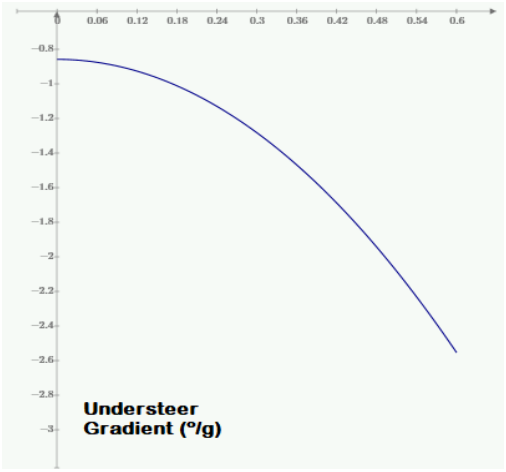
Figure 6: Front suspension sprung mass response to road displacement inputs as a function of



The suspension geometries were chosen with the aid of the AHP method (whose highest consistency ratio is 0.048, as defined by Saaty). Higher weights were assigned to lateral dynamics parameters, such as rolling center height and toe angle, due to their great influence on the understeer gradient. Higher values were also given to parameters which are easier to control, according to [11], [12] and the team's experience. The decision matrices showed that the fittest option for both rear and front suspensions was double wishbones.

Along with several minor kinematic goals the suspension hardpoints were chosen so as to produce a slightly oversteer vehicle, allowing it to negotiate sharper turns when needed. The minimum value requirement was set based on the driver's experience. The resulting understeer gradient (see Figure 7) is -0.9 °/g at 0.1g lateral acceleration, and -2.5 °/g at 0.6g (higher than the sought -3°/g).

Figure 7: Understeer gradient as a function of g's of lateral acceleration



Many obstacles and road profiles were sketched in multibody simulation software CarSim in order to better understand the loads acting on the vehicle in multiple scenarios. The critical values were then obtained and used as input in structural simulations in ANSYS containing all suspension components of a single wheel. The ball joints connections were modeled as spherical joints, and the pivot points of the wishbones as cylindrical joints. In the vertical load case, the tire was held in place, and the loads applied to the shock absorber so a transient analysis could be done, whereas in the combined vertical and lateral loads case the shock absorber was fixed and the loads applied to the tire. As to reduce computational time secondary components were modeled as rigid bodies.

The vertical load considered was taken from the vehicle response to a 2-meter fall, which had a peak vertical acceleration of 4.4 g's (forces acting on the springs and shock absorbers can be seen in Appendices **A13** and **A14**. In turn, the lateral load was drawn from the response to a high-speed side impact which resulted in a lateral acceleration of 1.9 g's. This case was of special interest because similar suspension builds had failed that way in the past, so a 1.5 factor safety requirement was established when designing the parts. Moreover, a fatigue analysis was conducted under those loads and all components reached infinite life.

2.6 FRAME / BODY / AERO

The structural frame of the project in question is based on the MB17 Model of the Minerva Baja car. In this session, the analyses are from this previous model as well. They are completely viable and relevant because the conditions endured at a standard Baja SAE competition are more demanding on the structure in comparison with the track utilized as reference in the challenge.

Due to time management, efficiency, and practical applications, a frame that is already built, analyzed and tested is a more viable decision in comparison with projecting a new one. Therefore, only the necessary adjustments to the hydrogen powertrain packaging are needed.

These adaptations concern the placement of the Electric Motor, Inverters, Converters, and the Stack Ballard FCgen®-1020ACS and all the components described in the segment 2.2) of this document.

Regarding the project of the frame done before the SAE Brazil & BALLARD Student H2 Challenge 2020, the following goals are set based on previous flaws:

- Wider structural and torsional stiffness studies with load conditions similar to track conditions in Baja SAE competitions.
- Enhance ergonomic studies in comparison to previous projects.
- Develop higher efficiency methods to choose materials and fabrications procedures.

Chassis – Two materials are considered to be utilized: SAE Steel 1020 and SAE Steel 4130. This choice is made from the Decision Matrix developed by the team, present at Table 7, in which the weights were defined according to the goals set for the project. From previous feedback, due to some misusing of the criteria in the matrix, some of the factors are reconsidered.

Table 7: Decision Matrix for chassis' tube material

Factors	Weight	SAE 4130		SAE 1020	
		Grade	Partial	Grade	Partial
Resilience	0,4	10	4,0	6	2,4
Tenacity	0,1	10	1,0	6	0,6
Mass	0,2	5	1,0	4	0,8
Fabrication/Repair Time	0,1	3	0,3	6	0,6
Cost	0,3	5	1,5	9	2,7
Total	1,1	7,8		7,1	

Decision Matrix Criteria:

1) Tenacity / directly related to Rupture Tension / [0(0Mpa) a 10(670(Mpa)]
2) Resilience / directly related to Yield limit / [0(0Mpa) a 10(550Mpa)]
3) Chassis Total Mass / [0(40kg) a 10(25kg)]
4) Total Cost = Chassis material costs + Fabrication and Welding costs / [0(2000 reais) a 10(0reais)]
5) Fabrication and Repair Time (in man-hour) / [0(160hh) a 10(0hh)]

Observation: Chassis Material Costs = related to price/kg of material + 25% of the value of leftovers and other expenses.

Table 8: Data from each material

Data	SAE 4130	SAE 1020
Primary Profile(mm)	31,75 x 1,6	25,4 x0,9
Secondary Profile(mm)	28,6 x 2	25,4 x0,9
Mass (Kg)	32,05	34,6
Price/weight(R$)	17,15	4,66
Yield Limit (Mpa)	550	350
Rupture Limit (Mpa)	670	440
Fabrication Cost(R$)	270	115
Fabrication Time(man-hour)	115	72

The tube profiles, fabrication templates of the chassis, weld type, pre and post processes of welding are quoted in reference [15].

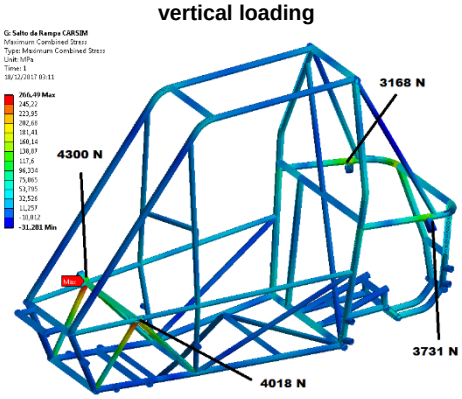
Structural Studies - It's extremely important to know the types of load applied in the chassis to evaluate its resistance and performance from other components. According to [16], there are various types of load in which 2 are principal: the ones that cause Longitudinal Torsion and the ones that cause Vertical Bending. Thus, analyses from these phenomena are made.

Vertical Loading – On BAJA SAE competitions, at the standard condition, the vehicle has to overcome obstacles such as holes, ramps, stones, tree trunks, etc. Based on [13], these forces at these conditions cause a maximum vertical acceleration of 5g, which has to be supported by the frame keeping a comfortable safety factor for dynamic loads.

To find the biggest load from obstacles the software CARSIM is utilized at various circuits similar to the competitions. The maximum load found in a situation similar to a real case is a ramp without reception at 36km/h, in which the first contact with the road is made by the front tires. This situation causes an acceleration of 4,5 g in the vertical direction, similar to the reference [13].

From this data, a Static Structural simulation is made with the use of Inertial Relief and Beam Elements to represent the dynamic situation. This command generates an acceleration that makes the resultant be zero, making fixations unnecessary. The forces at the front and rear follow the direction of the springs, and their magnitude are present at Figure 8. The maximum stress is 266,5 MPa, which results in a safety factor of 2,06. To reach a fine range of work stress it's necessary to make a lattice at the shock absorber pick up point, between the Fore-Aft Bracing frontal members and the Rear Roll Hoop members.

Figure 8: Forces and stress distribution for vertical loading



Torsional Stiffness – For the construction of the chassis, it's necessary to know the effects of Longitudinal Torsion on the frame. The deflection it may cause can affect negatively the handles of the vehicle due to changes in Lateral Load Transfer [14].

To surpass this issue, a well-based study about torsional stiffness is done in order to evaluate if the dynamic characteristics will be affected by the flexibility of the chassis. Therefore, this procedure is divided into 3 stages: acquisition of torsional stiffness experimentally, obtention of this property from a software CAE (Ansys), and analysis of the influence the chassis have in the lateral load transfer.

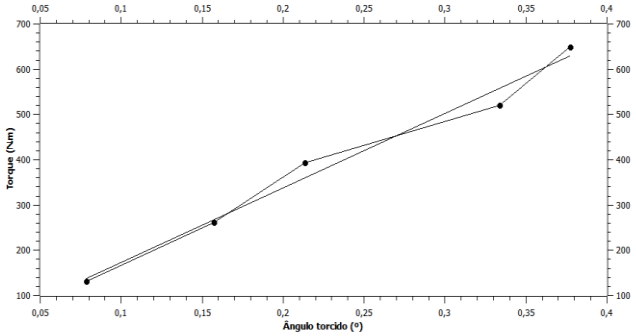
Experimental Torsional Stiffness – For the test, beams with a transverse section in the shape of I's are bonded with a leveled table at the rear. The shock absorbers are replaced by rigid bars, and the front is fixated on an I-shaped beam that is able to rotate as weight is added in an end of the beam. To evaluate the torsional stiffness, there must be torque (given by the lever arm and the weights) and

angular deflection (given by vertical deflections acquired with a dial indicator). The lower A-arm is directly connected with the vehicular dynamics due to the displacement of the hardpoints. Several measurements of torque and angular deflection are made and, by linear regression, the torsional stiffness is obtained along with the expanded uncertainty (due to linear regression, dial indicator, weights, etc.) (16±2)102 N.m/° (Figure 10)

Figure 9: Stiffness Workbench

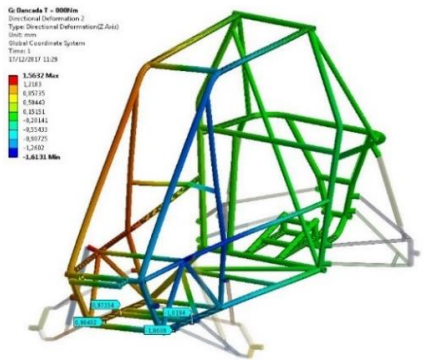


Figure 11: Obtaining the Torsional Stiffness via Linear Regression



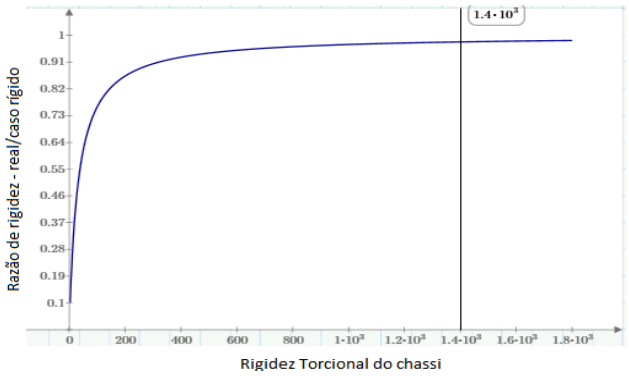
Torsional Stiffness from Finite Element Analysis – For this simulation, beam elements are used and boundary conditions are represented by the experimental conditions mentioned. Based on [16], fixed supports are applied at the rear and a binary is applied at the front. From this, similarly to the experimental analysis, the torsional stiffness of 1792 Nm/º (Figure 11) is obtained. Comparing the results, it's clear that they're compatible since the value is in the uncertainty range. The lack of accuracy in the results could be caused by the loosening in the rotations of the suspension components, reducing the torsional stiffness.

Figure 12: Torsional Stiffness FEA simulation



The paper [16] proposes a group of Frame-Suspension as an association of torsional springs in series, given that the front suspension is connected to the rear by the chassis. To calculate the equivalent stiffness of the system, the chassis stiffness and the roll stiffness of the front and rear suspensions are necessary (2940 e 4556 Nm/rad, respectively). Conservatively, the value of 1400Nm/º is chosen because it's the farthest value from the ideal rigid case. By the end of the equivalent stiffness value calculation, it can be noted that the torsional stiffness of the frame is 97% of the rigid case. This value is considered by literature as rigid and therefore it doesn't compromise vehicular dynamics.

Figure 10: Convergence of the flexible model



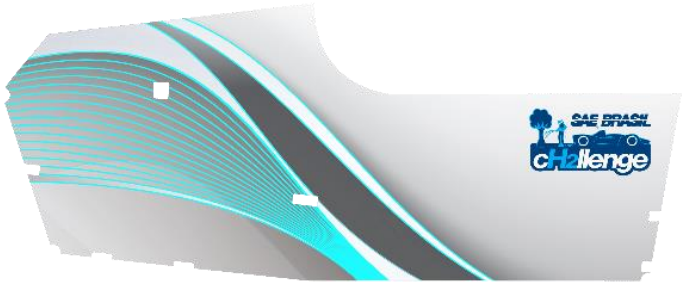
Body – The main objective of this project's design is related to the topic "Hydrogen for Brazilian Mobility". "Green"; "Eco-Friendly"; "Sustainable Energy"; "Clean Energy" and "Hydrogen". These are the primary words and concepts that must relate to this car's Body and Shape. Secondly, the concepts "Water" and "Flow" were thought, as it has a sense of dynamics, and resembles how the world is shifting smoothly to a more conscient environment. In addition, water is the main source of hydrogen, thus indispensable to our design.

First, the design team gathered inspirations for color and shapes. Those that stood up were the Mercedes F1 Car with a contrast of grey and black; waves, and water flows. As the primary color choice of Cyan Green, as the mix of green from the premise of an eco-friendly vehicle, and blue from the water as it's the main source of hydrogen.

The Cowl is the same format as previous ones utilized in this car. Considering that viability to build the car within the period stipulated is a criterion of scoring, the team chose to minimize risk, time, and money in a totally innovative shape. Furthermore, the cowl with few split parts and lateral plates fit adequately our purpose of design, with the capability of showing a flow without breaking the lines with physical divisions.

As a bigger change in shape, a rear cowl has been implanted to protect the powertrain from direct sunlight and overheating.

Figure 13: Vehicle Livery Design



2.7 ELECTRONICS / ELECTRICAL –

In order to increase the driver's performance and ensure the monitoring of the car's vital signs, in addition to meeting market expectations and improving the performance of other systems, an embedded system and a panel for the display of information have been developed.

The panel has as objectives the display of information such as speed, distance, battery voltage, hydrogen level and general warning messages.

OVERVIEW - The embedded system has its own key that allows it to be turned off, in order to increase the battery's autonomy, because in some moments, as in a punishment during the endurance test of Baja SAE, the car remains parked, and commercial speaking, the costumer may bring the car to a halt and not want it to keep draining energy. The onboard electronics are powered by the LM2596 voltage regulator, which has low heat emission and a wide voltage regulation range that can be adjusted. Based on the current generated at the startup of the motor, the ideal cross-sectional area for the wires is 1.5 mm². While for the electronic part, wires of 0.5 mm² cross-sectional area are used due to the low current required.

POWER SUPPLY - The supply of the embedded electronics is independent from the traction power supply. It is made by an auxiliary 12V battery and the voltage regulator LM2596, with the presence of an additional 1A fuse to protect the sensors and microcontroller. The fan, on the contrary, shall be powered by the traction power supply, given its high relative voltage and sum importance to the proper functioning of the system.

PROCESSMENT - In this project, aiming for innovation as a whole, the ESP32 LORA microcontroller is used. A large performance gain is planned in the development of a sensing and telemetry system coupled to the double-sided plate. The referred microcontroller has a high processing capacity, low energy consumption, a large number of analog and digital ports and allows easy programming in C / C ++ language. In addition to possible connections via Wi-fi, Bluetooth and the Lora network integrated in the microcontroller that allows a more compact and efficient system, it also allows a large signal transmission range of up to 4km, no need to use other MCUs. The ESP32 has two processing cores and a clock frequency of up to 240MHz with several inputs that facilitates the expansion of the project in the vehicle data acquisition.

COMPETITION-ORIENTED SENSING

Speed - The speed of the prototype is estimated from the average speed of the front wheels; Hall effect sensors (DRV5053) are able to detect the passage of magnets attached to them. With the amount of 6 magnets in each wheel and a measurement frequency of 1Hz, an accuracy of about 1km / h is achieved, also measuring the speed of the rear wheels it is possible to know when the car is sliding. The distance is calculated by the product between the circumference and the average turn of the wheels and is saved on an SD memory card. A logic signal warns the micro-controller when the power is stopped and a capacitor ensures that it remains on for 3.3ms so that the rescue is done.

Telemetry - For the transmission of the data captured by the embedded system, the Lora communication integrated in the ESP32 LORA microcontroller is used. It has a low energy consumption, low cost and that reaches long communication distances of up to 4km with a power of 20dBM or 100mW that allows you to efficiently and dynamically monitor important car parameters even over long distances. For short-range or static tests and validations, WiFi or Bluetooth BLE (Bluetooth Low Energy) is also used, which is integrated in the microcontroller that still allows interaction with the car's embedded electronics with computers or cell phones that have internet access.

Hydrogen level (H2) - The hydrogen measurement system is performed by a stainless steel OEM pressure sensor that withstands pressures up to 3000 psi specific for hydrogen mobile hydraulics applications. The model is the MH-3-HY and is installed before the pressure regulator. Because with the pressure variation we estimate the volume of hydrogen contained in the cylinder to indicate the autonomy on the vehicle's dashboard. An LCD display is used to dis-play this information to the pilot.

Battery current voltage – A digital DC Ammeter and a voltmeter are connected on the battery terminals alongside the BMS (Battery Management System), which allows the possibility to measure voltage and current, besides power and energy.

Panel - The panel was designed to show, in a simple way, information about the car that could influence the driver's decisions. An LCD display and LEDs, which make up a warning system, were used to display information. The LCD display shows the speed, hydrogen level, battery voltage and current, temperature of the fuel cell.

Warning system - Thinking about the safety of the pilot and the maintenance of the prototype, a warning system is placed on the panel, warning the pilot by means of LEDs when there is a low volume of hydrogen, a dead battery or whenever a certain distance is covered.

Cost - Aiming at a higher project feasibility, all the components are chosen with the highest possible cost-benefit ratio. It is listed below the average cost of the main components found in the market.

Table 92: Electronic Components Cost

Component	Price
ESP32 LORA	R$ 60,00
Hall effect sensor	R$ 7,00
Voltage regulator	R$ 12,00
H2 pressure sensor	R$ 300,00
DC voltimeter	R$ 50,00
DC amperimeter	R$ 50,00
LCD display	R$ 30,00
Total	**R$ 509,00**

2.8 – Hydrogen Security System

2.9 – Vehicle's Performance Theoretical Evaluation

OBJECTIVE - It becomes essential to evaluate the vehicle's performance and some other crucial parameters such as its energy and fuel consumption, range, lap times, powertrain efficiency and other data. For that, OptimumLap was used to run a preliminary simulation, and then ADVISOR (Advanced Vehicle Simulator), a MatLab based open-source software developed by NREL, was used to run a thorough simulation that provided deeper and more accurate information on efficiency, fuel consumption, and the overall powertrain behavior through the defined cycle. As a helping guide to our computational evaluations we used [24], which was oriented by UFRJ Professor Sílvio Carlos Aníbal, who is also guiding us through this project alongside our Advisor and some of the UFRJ LABH2 scientists responsible for the development and implementation of Fuel-Cell based buses in our Campus.

PRELIMINARY SIMULATION – OptimumLap was previously used to determine the ideal chain reduction through an iterative analysis, as shown in 2.3 , so, after altering the input data to fit the ideal reduction of 13.7:1, the simulation was run using the provided circuit as its cycle. As a result, the Speed x Time graph [**A15**], alongside its CVS table was outputted. That information shall be used to define the drive cycle on ADVISOR later on, but the data contained in the .CVS wasn't treated, and thus couldn't be read nor comprehended by ADVISOR. In that sense, we passed it through a simple Excel iterative data treating scheme the team developed, successfully making it readable.

INPUT DATA ON ADVISOR –

Preliminary Data - The software already has a database that can be used as base parameters, selecting the best fitting components which are most similar to those used in this project, then altering specific information to specify our project. The base choices are described below:

Table 10: ADVISOR base data choices

Parameter	Choice	Reason
Drivetrain Configuration	FUEL_CELL	Defines a fuel cell hybrid vehicle.
Vehicle	VEH_SMCAR	Specifies vehicle data according to a small vehicle. especially useful in estimating non-available data.
Energy Storage	ESS_LI7_TEMP	Only option describing a Lithium-Ion Battery.
Motor	MC_AC25	Option describing an AC induction motor with the smallest power.
Transmission	TX_1SPD	Describes a one speed transmission. Rear wheel drive was also selected
Wheel/Axle	WH_TractorTrailer_lowRR_SS	Option most similar to our available Baja tires
Powertrain Control	PTC_FUELCELL	Describes a common control system for a FC hybrid vehicle.

Input Data – Regarding the vehicle as a whole, the following parameters were defined:

a) Drag Coefficient: 0.29, defined through a CFD analysis as shown in **A13**;

b) Frontal Area: 0.88 m²

c) Vehicle Mass: 315kg, as estimated previously;

d) Vehicle Center of Gravity Height: 0.54m, as per a CG experiment (shown in **A16**) done on the Baja platform before the adaptations, considered to be a good estimative;

e) Wheelbase: 1.5m

Regarding the wheel, the following parameter were defined:

a) Wheel Radius: 0.28m

b) Wheel Inertia: 1.2304

c) Rolling Resistance Coefficient: 0.03, defined through **A1** and validated through a coast down test, based on [5] and [6].

The battery characteristics depends on its working temperature, ADVISOR interpolates those characteristics on three different temperatures to the Lithium-Ion batteries. Appendix **A18** shows the temperature effects on battery power. And after simulating the battery temperature after 15 laps on the reference circuit, we were able to determine the average working temperature after stabilization to be accurately near 41ºC, so we used data from that curve. Finally, having in mind all the previous information, the following parameters were defined, considering the use of the LG Chem 18650 HD2 Cell:

a) Maximum and Minimum Voltage: 2.5 and 4.2V, according to the chosen battery's datasheet. That data is referring to each of its 14 modules, which was also described in the program;

b) Cap Scale: 0.24, as a generic Lithium-Ion battery was selected as base, then was determined a scale value that would alter the base parameters (mass, module power and module capacity) accordingly in order to fit the desired values.

c) State of Charge: From the data in **A18**, having in mind the goal to extend the battery pack life, we determined the minimum state of charge (SOC) as being 40% of its full charge, because of a significant drop in the instantaneous power for SOC's below that and also to protect it from peak discharges. The maximum SOC was determined as 0.8 of its maximum, in order to protect the battery from recharge peaks during regenerative braking when the pack may be already almost fully charged. Furthermore, the initial SOC was defined as the average SOC during the defined drive cycle. That was done to minimize the influence of the acceleration lap on the fuel consumption output data, as the first lap naturally consumes more energy from the battery than from the FC.

As for the Fuel Cell, the following data was inputted to adapt the ADVISOR model to our own:

a) Fuel cell maximum power: 2.1kW

b) Power rise rate: set as 420 W/s, in order for the FC to reach its maximum power in 5 seconds, as recommended by [22]

c) FC minimum power: 300W. That value was determined due to a heavy loss in efficiency for operating powers below that point (as can be seen in appendix **A19**), due to system accessories such as the cooling fan.

d) It was also chosen to use the Charge Sustain Mode, instead of the Charge deplete, in order for the hybridization rate to favor the fuel cell instead of the battery pack. In this operation mode the battery Works only on accelerations, on the FC low-efficiency ranges, during energy recovery while braking and when the FC doesn't meet the full power required by the control system.

As we had the fuel tank volume (5.9L) and working pressure (153 bar), through a simple calculation the total maximum fuel mass was determined to be 0.066kg and its density as 11.18 g/l.

RESULTS –

3. CONCLUSION

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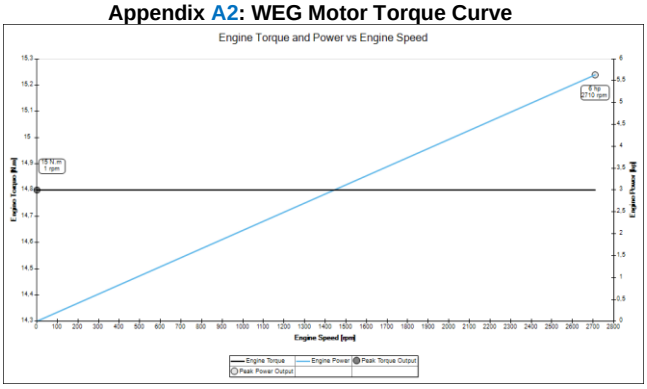
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5. APPENDIX

Appendix A1: Rolling Resistance Experiment



Appendix A2: WEG Motor Torque Curve

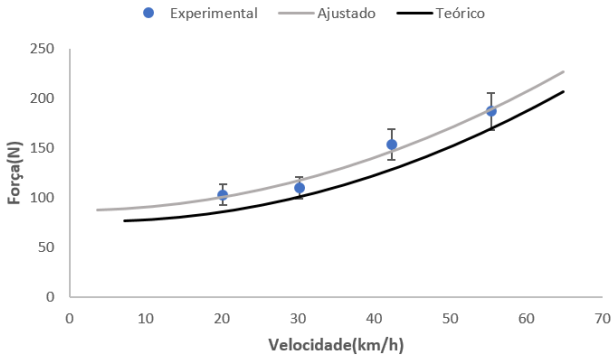


Appendix A3: CFD Analysis in Solidworks

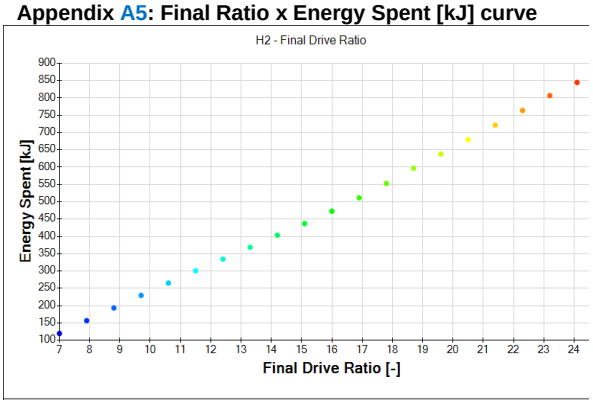


CFD ANALYSIS TO OBTAIN DRAG COEFFICIENT

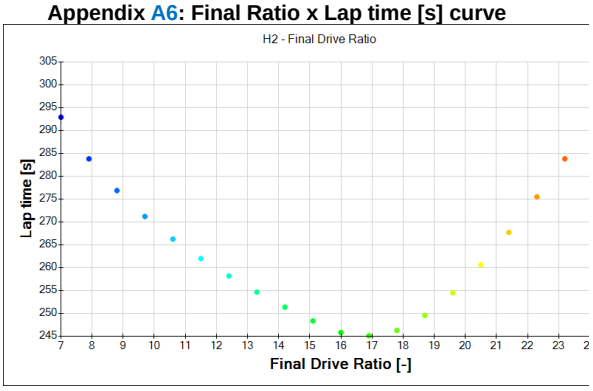
Appendix A4: Comparison between experimental and theoretical data of the resistive forces



Appendix A5: Final Ratio x Energy Spent [kJ] curve



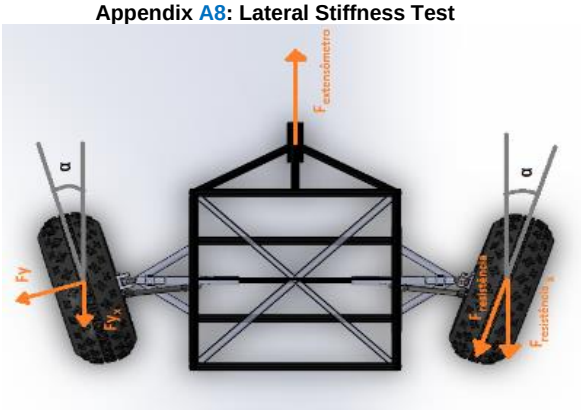
Appendix A6: Final Ratio x Lap time [s] curve



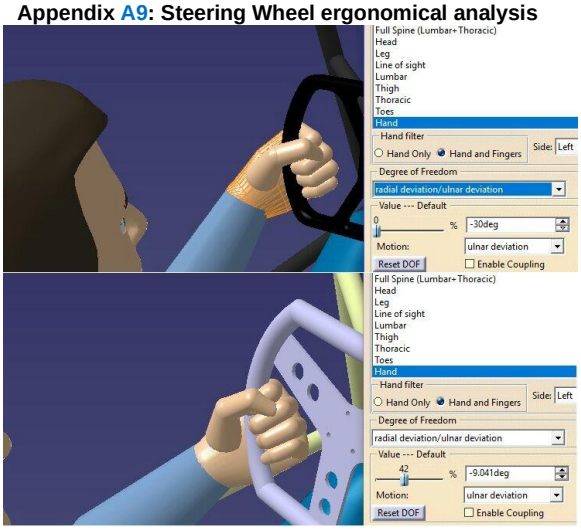
Appendix A7: LFD-15 Pressure Regulation Valve



Appendix A8: Lateral Stiffness Test



Appendix A9: Steering Wheel ergonomical analysis



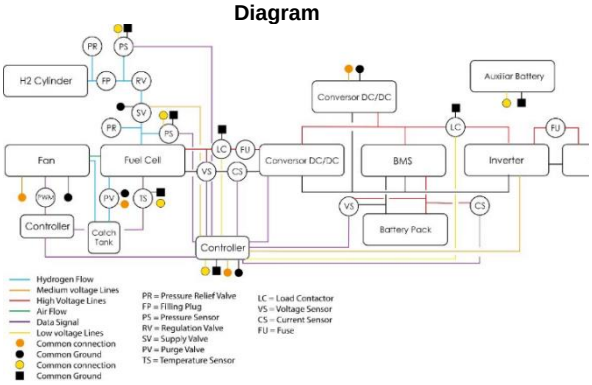
Appendix A10: Comparison between possible brake disc materials

Materiais	Fe cinzento	Fe Nodular	SAE 1045	SAE 1020	Inox 420
Limite de escoamento (Mpa)	-	379	530	350	345
Condutividade térmica (W/m*K)	46	36	51,9	51,9	30
Calor específico (J/kg*K)	544	544	486	486	460
Densidade(kg/m³)	7300	7100	7860	7870	7700

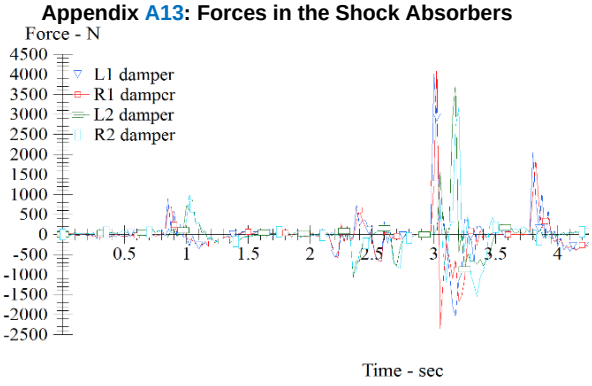
Appendix A11: Inverter WEG CVW 300



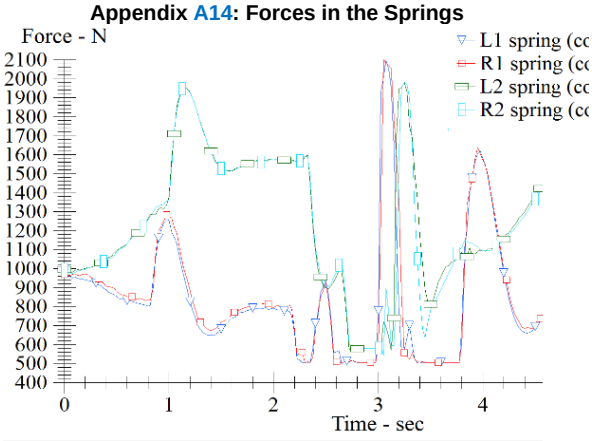
Appendix A12: Powertrain Components Block Diagram



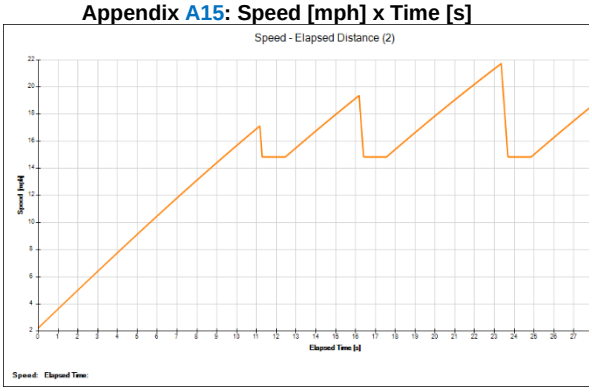
Appendix A13: Forces in the Shock Absorbers



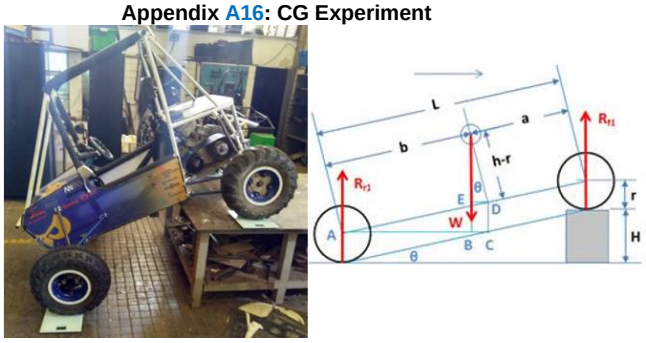
Appendix A14: Forces in the Springs



Appendix A15: Speed [mph] x Time [s]



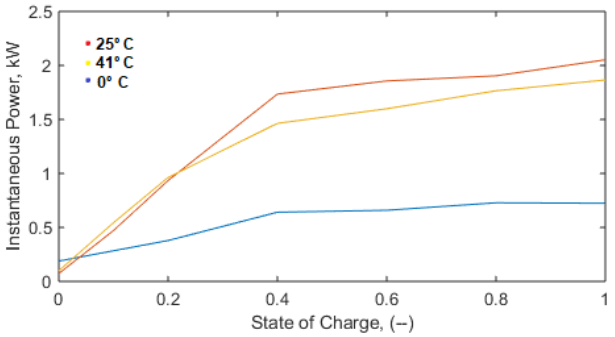
Appendix A16: CG Experiment



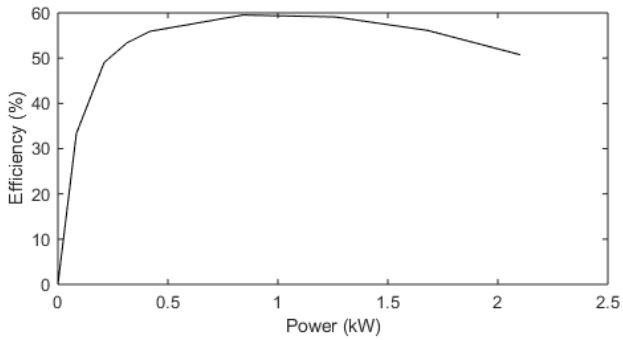
Appendix A17: H2 Cylinder Cradle



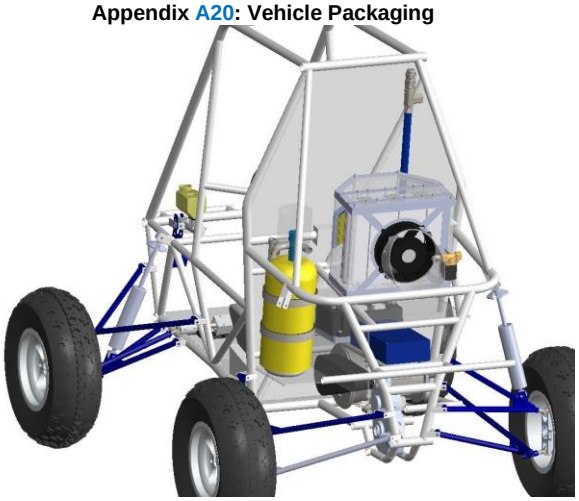
Appendix A18: Battery Pack instantaneous power [kW] x SOC for three working temperatures



Appendix A19: Fuel Cell efficiency [%] x Power [kW]



Appendix A20: Vehicle Packaging



Appendix A21: Fuel Cell Enclosure

